

Dwell-Conditioned Cycle-to-Cycle Switching-Time Feedback for Mode-Dependent Switched-Affine Systems

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Abstract

Fixed-period switching-time feedback can regulate a reference periodic trajectory with a static cycle-to-cycle law, but an unconstrained law can request negative or otherwise inadmissible dwell durations. This paper presents an implementation-oriented feasibility layer for such feedback: the complete raw switching-instant-offset vector is multiplied by the largest scalar in $[0, 1]$ that satisfies every minimum dwell duration. The conditioner preserves the requested direction and cycle period, requires only an $O(N)$ scan for an N -interval schedule, and invokes no online solver. For the linearized cycle model, conditioning produces the segment $A(\beta) = (1 - \beta)\Phi + \beta A_{c1}$. A common quadratic Lyapunov certificate at its endpoints proves contraction for arbitrary conditioning-factor sequences. As an enabling foundation, a fixed-period one-cycle model is derived by exact augmented matrix-exponential propagation for a prescribed sequence of affine modes whose state matrices and affine vectors may both change. This derivation recovers the established switching-sensitivity structure and the narrower switched-actuator model as a special case. On a nine-interval converter benchmark, analytical Jacobians agree with central differences to relative errors of $8.518\text{e-}12$ and $8.344\text{e-}11$, and the residual slope is 2.000. From a large initial error, the raw aggressive law requests a $-43.045\text{ }\mu\text{s}$ dwell; conditioning enforces the assumed $3\text{ }\mu\text{s}$ bound and outpaces a feasible conservative law in exact nonlinear simulation. The Lyapunov result is for the linearized model; nonlinear conclusions remain local.

Keywords: switched-affine systems; periodic trajectories; switching-time control; dwell-time conditioning; one-cycle map; DC–DC converter.

1 Introduction

Switched-affine systems evolve through a finite collection of affine vector fields. Under a periodic switching schedule, the desired steady behavior is generally a reference periodic trajectory rather than a static equilibrium. This work fixes the nominal mode sequence and cycle period and regulates that trajectory by moving only the interior switching instants. The practical difficulty is coupled feasibility: moving one instant lengthens one dwell duration and shortens its neighbor, so an unconstrained feedback action can invert an interval even when each offset appears individually moderate.

The cycle-model foundation builds on published prior art. Patiño, Riedinger, and Ruiz derived analytical switching-instant sensitivities for cyclic systems with mode-dependent dynamics and used them in online Gauss–Newton predictive control [1]. Marcolino, Galvão, and Kienitz derived a fixed-period one-cycle model and discrete LQR law for an LTI plant with piecewise-constant actuator levels, together with dwell inequalities and constrained predictive control [2]. Thus neither switching sensitivity, fixed-period timing coordinates, nor static LQR is claimed here as a new control principle.

The augmented derivation below provides a controller-ready connection between the mode-dependent sensitivity and the narrower switched-actuator formulation.

The paper instead centers on a constraint-handling and certification layer for existing one-cycle feedback. Given any raw switching-instant action, a scalar conditioner applies the largest feasible multiple of the entire vector. Unlike componentwise clipping, this radial operation preserves the raw direction and the two fixed cycle boundaries. Its scalar is available by inspecting the N dwell changes, without online optimization. A common quadratic Lyapunov test then certifies the full linear family generated while the conditioner is active.

The contributions are:

1. a direction-preserving, maximal feasible dwell-time conditioner with an explicit $O(N)$ formula and raw-action admissible region;
2. a common-Lyapunov certificate proving contraction of the conditioned linearized model for arbitrary state-dependent scaling sequences; and
3. supporting technical results: a compact augmented-state derivation in independent timing coordinates, its specialization to the switched-actuator model, direct Jacobian and residual validation, and an exact nonlinear converter study.

The first two items are the central contribution. The derivation clarifies and validates the relationship between the published mode-dependent sensitivity and switched-actuator formulations; LQR supplies raw comparison laws rather than a new synthesis method. No worldwide-priority or global nonlinear-stability claim is made.

2 Fixed-Period One-Cycle Model

2.1 Switched-affine cycle and timing coordinates

Consider

$$\dot{\mathbf{x}}(t) = A_{\sigma(t)}\mathbf{x}(t) + \mathbf{b}_{\sigma(t)}, \quad \mathbf{x}(t) \in \mathbb{R}^n, \quad (1)$$

where one cycle follows the fixed sequence $\boldsymbol{\Omega} = (\omega_1, \dots, \omega_N)$ over nominal boundaries

$$0 = \bar{t}_0 < \bar{t}_1 < \dots < \bar{t}_N = T, \quad \bar{d}_i = \bar{t}_i - \bar{t}_{i-1}. \quad (2)$$

The nominal schedule generates a reference periodic trajectory $\bar{\mathbf{x}}(t)$. Its cycle anchor $\mathbf{x}^* = \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}(T)$ defines the cycle-start error $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$.

Only the interior switching instants move:

$$t_{k,i} = \bar{t}_i + \delta\tau_{k,i}, \quad i = 1, \dots, N-1, \quad \delta\tau_{k,0} = \delta\tau_{k,N} = 0. \quad (3)$$

Collect the $N-1$ independent offsets in $\delta\boldsymbol{\tau}_k$. Their N dependent dwell changes satisfy

$$\delta\mathbf{d}_k = D\delta\boldsymbol{\tau}_k, \quad D_{ij} = \begin{cases} 1, & i = j, \\ -1, & i = j+1, \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

for $i = 1, \dots, N$ and $j = 1, \dots, N-1$. Hence $\delta d_{k,i} = \delta\tau_{k,i} - \delta\tau_{k,i-1}$ and $\mathbf{1}^\top D = 0$, so every offset vector preserves T . The exact one-cycle map is $\mathbf{x}_{k+1} = \mathcal{P}(\mathbf{x}_k, \delta\boldsymbol{\tau}_k)$.

2.2 Exact augmented propagation

Introduce the augmented state, its physical-state projection, and the mode generators

$$\mathcal{X} = \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}, \quad \Pi = [I_n \quad \mathbf{0}], \quad F_i = \begin{bmatrix} A_{\omega_i} & \mathbf{b}_{\omega_i} \\ \mathbf{0}^\top & 0 \end{bmatrix}. \quad (5)$$

The affine interval dynamics become $\dot{\mathcal{X}} = F_i \mathcal{X}$. For duration s , define $E_i(s) = e^{F_i s}$ and the nominal transition $\phi_i = E_i(\bar{d}_i)$. Since $d_{k,i} = \bar{d}_i + \delta d_{k,i}$, exact propagation over one cycle is

$$\mathcal{X}_{k+1} = E_N(\bar{d}_N + \delta d_{k,N}) \cdots E_1(\bar{d}_1 + \delta d_{k,1}) \mathcal{X}_k, \quad (6)$$

where the first interval is the rightmost factor. Let

$$\bar{\mathcal{X}}_i = \phi_i \cdots \phi_1 \bar{\mathcal{X}}_0, \quad \tilde{\mathbf{e}}_k = \begin{bmatrix} \mathbf{e}_k \\ 0 \end{bmatrix}. \quad (7)$$

Periodicity gives $\phi_N \cdots \phi_1 \bar{\mathcal{X}}_0 = \bar{\mathcal{X}}_0$. Subtracting this nominal cycle from (6) yields the exact augmented terminal error

$$\begin{aligned} \tilde{\mathbf{e}}_{k+1} &= E_N(\bar{d}_N + \delta d_{k,N}) \cdots E_1(\bar{d}_1 + \delta d_{k,1}) \\ &\quad \times (\bar{\mathcal{X}}_0 + \tilde{\mathbf{e}}_k) - \phi_N \cdots \phi_1 \bar{\mathcal{X}}_0. \end{aligned} \quad (8)$$

2.3 First-order expansion in dwell durations

For constant F_i , differentiation with respect to duration gives

$$\frac{d}{ds} e^{F_i s} = F_i e^{F_i s}, \quad (9)$$

so each perturbed interval transition satisfies

$$E_i(\bar{d}_i + \delta d_{k,i}) = \phi_i + F_i \phi_i \delta d_{k,i} + \mathcal{O}(\delta d_{k,i}^2). \quad (10)$$

Substituting (10) into the ordered product and retaining terms containing at most one dwell perturbation gives the following expression, where E_i on the left denotes $E_i(\bar{d}_i + \delta d_{k,i})$:

$$\begin{aligned} E_N \cdots E_1 &= \tilde{\Phi} + \sum_{i=1}^N \phi_N \cdots \phi_{i+1} F_i \phi_i \cdots \phi_1 \delta d_{k,i} \\ &\quad + \mathcal{O}(\|\delta \mathbf{d}_k\|^2), \quad \tilde{\Phi} = \phi_N \cdots \phi_1, \end{aligned} \quad (11)$$

where an empty product is the identity. Inserting this expansion into (8) cancels the nominal term. Products $\delta d_{k,i} \tilde{\mathbf{e}}_k$ are second order and are discarded. Since $\phi_i \cdots \phi_1 \bar{\mathcal{X}}_0 = \bar{\mathcal{X}}_i$, the remaining first-order terms are

$$\tilde{\mathbf{e}}_{k+1} = \tilde{\Phi} \tilde{\mathbf{e}}_k + \sum_{i=1}^N \tilde{\mathbf{g}}_i \delta d_{k,i} + \mathcal{O}\left(\left\| \begin{bmatrix} \mathbf{e}_k^\top & \delta \mathbf{d}_k^\top \end{bmatrix} \right\|^2\right), \quad (12)$$

with

$$\tilde{\mathbf{g}}_i = \phi_N \cdots \phi_{i+1} F_i \bar{\mathcal{X}}_i. \quad (13)$$

Projection onto the physical state gives

$$\mathbf{e}_{k+1} = \Phi \mathbf{e}_k + G_d \delta \mathbf{d}_k + \mathcal{O}\left(\left\| \begin{bmatrix} \mathbf{e}_k^\top & \delta \mathbf{d}_k^\top \end{bmatrix} \right\|^2\right), \quad (14)$$

where

$$\Phi = e^{A_{\omega_N} \bar{d}_N} \cdots e^{A_{\omega_1} \bar{d}_1}, \quad G_d = \Pi [\tilde{\mathbf{g}}_1 \cdots \tilde{\mathbf{g}}_N]. \quad (15)$$

2.4 Conversion to switching-instant offsets

Using $\delta \mathbf{d}_k = D\delta \boldsymbol{\tau}_k$ in (14) produces the controller-ready model

$$\mathbf{e}_{k+1} = \Phi \mathbf{e}_k + \Gamma_\tau \delta \boldsymbol{\tau}_k + \mathcal{O}\left(\left\|[\mathbf{e}_k^\top \delta \boldsymbol{\tau}_k^\top]\right\|^2\right), \quad \Gamma_\tau = G_d D. \quad (16)$$

Column i of D adds dwell i and subtracts dwell $i+1$, hence $\Gamma_{\tau,i} = \Pi(\tilde{\mathbf{g}}_i - \tilde{\mathbf{g}}_{i+1})$. Because F_{i+1} commutes with $\phi_{i+1} = e^{F_{i+1}\bar{d}_{i+1}}$, adjacent terms can be factored to obtain

$$\begin{aligned} \Gamma_{\tau,i} &= e^{A_{\omega_N}\bar{d}_N} \dots e^{A_{\omega_{i+1}}\bar{d}_{i+1}} \\ &\times \left[(A_{\omega_i} - A_{\omega_{i+1}})\bar{\mathbf{x}}(\bar{t}_i) + \mathbf{b}_{\omega_i} - \mathbf{b}_{\omega_{i+1}}\right]. \end{aligned} \quad (17)$$

The bracket is the jump between adjacent affine vector fields. This is the established switching-sensitivity structure [1], now reached explicitly from duration perturbations and the independent fixed-period timing coordinates.

If every interval instead shares $A_{\omega_i} = A_c$ and $\mathbf{b}_{\omega_i} = B_c \mathbf{u}_{i-1}$, then

$$\Phi = e^{A_c T}, \quad \Gamma_{\tau,i} = e^{A_c(T-\bar{t}_i)} B_c (\mathbf{u}_{i-1} - \mathbf{u}_i), \quad (18)$$

which recovers Marcolino's switched-actuator timing model up to the input-jump sign convention [2]. Mode-dependent A_{ω_i} makes the jump state dependent and prevents that reduction in general.

3 Solver-Free Dwell-Time Conditioning

Let a static law request the raw switching-instant offset

$$\delta \boldsymbol{\tau}_k^{\text{raw}} = -K \mathbf{e}_k, \quad A_{\text{cl}} = \Phi - \Gamma_\tau K. \quad (19)$$

For required minimum dwell duration $d_{\min}$, assume the nominal margin

$$\mathbf{m} = \bar{\mathbf{d}} - d_{\min} \mathbf{1} > \mathbf{0}. \quad (20)$$

The raw-action admissible region is the polyhedron

$$\mathcal{C} = \{\mathbf{e} : \bar{\mathbf{d}} - D K \mathbf{e} \geq d_{\min} \mathbf{1}\}. \quad (21)$$

Inside $\mathcal{C}$, the raw action is already feasible. Outside it, define $\mathbf{q}_k = D\delta \boldsymbol{\tau}_k^{\text{raw}}$ and

$$\beta_k = \min \left\{ 1, \min_{i: q_{k,i} < 0} \frac{m_i}{-q_{k,i}} \right\}, \quad \delta \boldsymbol{\tau}_k^{\text{app}} = \beta_k \delta \boldsymbol{\tau}_k^{\text{raw}}, \quad (22)$$

where the inner minimum over an empty set is $+\infty$. The scalar β_k is the conditioning factor.

Proposition 1 (Feasibility and maximality). *Under (20), (22) returns the largest $\beta \in [0, 1]$ for which every applied dwell duration satisfies $\bar{\mathbf{d}} + D(\beta \delta \boldsymbol{\tau}_k^{\text{raw}}) \geq d_{\min} \mathbf{1}$. It preserves the cycle period and the raw-action direction, and it is computable in $O(N)$ time without an online solver.*

Proof. For $q_{k,i} \geq 0$, scaling cannot reduce dwell i . For $q_{k,i} < 0$, feasibility is equivalent to $\beta \leq m_i/(-q_{k,i})$; intersecting these bounds with $[0, 1]$ yields (22) and proves maximality. Also $\mathbf{1}^\top D = 0$ preserves T , while multiplication of the complete vector by one nonnegative scalar preserves its direction. One pass over the N dwell changes evaluates all bounds. $\square$

For the linearized model, the applied action gives

$$\mathbf{e}_{k+1} = A(\beta_k)\mathbf{e}_k, \quad A(\beta) = (1 - \beta)\Phi + \beta A_{\text{cl}}. \quad (23)$$

Thus active conditioning does not introduce an arbitrary saturated gain: it moves on the segment joining the nominal and raw closed-loop matrices.

Proposition 2 (Common-Lyapunov contraction). *Let $P = P^\top > 0$, $\|\mathbf{x}\|_P = (\mathbf{x}^\top P \mathbf{x})^{1/2}$, and suppose*

$$\|\Phi\|_P \leq q_0 < 1, \quad \|A_{\text{cl}}\|_P \leq q_1 < 1. \quad (24)$$

Then every $A(\beta)$, $\beta \in [0, 1]$, contracts in the same norm by at most $q = \max(q_0, q_1) < 1$. Consequently, the linearized conditioned model is exponentially stable for every sequence $\{\beta_k\} \subset [0, 1]$, including state-dependent sequences generated by (22).

Proof. The induced-norm triangle inequality gives $\|A(\beta)\|_P \leq (1 - \beta)q_0 + \beta q_1 \leq q < 1$. Therefore $\|\mathbf{e}_k\|_P \leq q^k \|\mathbf{e}_0\|_P$ for any sequence. $\square$

A common P may be sought offline through endpoint Lyapunov inequalities, which are standard semidefinite constraints [3]. This offline certificate is not online controller computation: online execution consists only of the matrix–vector raw law, $D\delta\boldsymbol{\tau}^{\text{raw}}$, and the scalar scan in (22).

The exact nonlinear conclusion is deliberately narrower. If $\mathcal{P}$ is continuously differentiable while mode order is preserved and A_{cl} is Schur, then strict nominal margin and $\delta\boldsymbol{\tau}^{\text{raw}}(\mathbf{0}) = \mathbf{0}$ imply that a neighborhood of the anchor lies in $\mathcal{C}$. The conditioner is inactive there ($\beta = 1$), so the exact closed-loop map has Jacobian A_{cl} and locally exponentially stabilizes the cycle anchor and its phase-locked reference periodic trajectory. Proposition 2 certifies arbitrary conditioning for the linearized model; it does not establish global stability of the nonlinear plant.

4 Converter Case Study

4.1 Three-cell converter benchmark and implementation

The case study uses the three-cell multilevel DC–DC converter reported by Patiño et al. [1], with state $\mathbf{x} = [v_{C_1}, v_{C_2}, i_L]^\top$. For binary switch commands $u_1, u_2, u_3 \in \{0, 1\}$,

$$A_{\mathbf{u}} = \begin{bmatrix} 0 & 0 & (u_2 - u_1)/C_1 \\ 0 & 0 & (u_3 - u_2)/C_2 \\ (u_1 - u_2)/L & (u_2 - u_3)/L & -R/L \end{bmatrix}, \quad \mathbf{b}_{\mathbf{u}} = \begin{bmatrix} 0 \\ 0 \\ Eu_3/L \end{bmatrix}. \quad (25)$$

Both terms change with the switch state, so the plant lies outside the common-state-matrix class of (18).

The reported physical mode sequence is $(0, 1, 3, 7, 2, 0, 4, 7, 4)$. The rounded cycle boundaries are $(0, 66, 88, 110, 132, 154, 220, 242, 264, 286)$ μs , and the reported cycle anchor is $\mathbf{x}^* = (9.9247, 19.2928, 0.9823)^\top$. Because the rounded data do not close exactly under exact propagation, a one-time least-squares reconciliation of the dwell vector enforces the published 286 μs period, anchor, and 22 μs nominal-design bound. It yields

$$\bar{\mathbf{t}} = (0, 63.889, 87.557, 109.557, 132.538, 154.538, \\ 218.210, 240.380, 263.828, 286) \mu\text{s}. \quad (26)$$

The largest boundary correction is 2.111 μs and the infinity-norm closure error is 1.421e-14. This reconciliation is data preparation, not a trajectory-design contribution.

Table 1: Converter, schedule, and feedback parameters. The $3\mu\text{s}$ applied bound is a simulation assumption distinct from the nominal benchmark-design bound.

Quantity	Symbol	Value
Source; capacitances	$E; C_1, C_2$	30 V; 40 μF
Inductance; resistance	$L; R$	10 mH; 10 Ω
Intervals; period	$N; T$	9; 286 μs
Nominal-design dwell bound	d_{bench}	22 μs
Applied-schedule dwell bound	d_{min}	3 μs
State; timing scale	$S_x; t_s$	$\text{diag}(10, 20, 1)$; 10 μs
Aggressive LQR weights	$Q_n; R_n$	I_3 ; $0.001I_8$
Conservative LQR weights	$Q_n; R_n$	I_3 ; $0.1I_8$

The cycle averages in figure 1 are 10.410 56 V, 19.422 28 V, and 0.956 13 A. The controller uses normalized coordinates $\mathbf{z} = S_x^{-1}\mathbf{e}$ and $\mathbf{v} = \delta\boldsymbol{\tau}/t_s$. The aggressive raw law is a standard discrete LQR with $Q_n = I_3$ and $R_n = 0.001I_8$; the conservative comparator uses $R_n = 0.1I_8$. Only the aggressive law is conditioned.

4.2 Linearization and certificate checks

Central differences of the exact one-cycle map use state steps equal to 10^{-4} times the diagonal entries of S_x and timing step $10^{-3}t_s = 0.01\mu\text{s}$. A second test applies simultaneous state and timing perturbations of normalized amplitude ϵ and evaluates

$$r(\epsilon) = \|S_x^{-1}[\mathcal{P}(\mathbf{x}^* + \Delta\mathbf{x}, \delta\boldsymbol{\tau}) - \mathbf{x}^* - \Phi\Delta\mathbf{x} - \Gamma_\tau\delta\boldsymbol{\tau}]\|_2. \quad (27)$$

The Jacobian errors and quadratic residual slope reported in table 2 directly validate the implemented first-order model. The complete analytical and finite-difference matrices and the residual sweep are retained in the generated numerical data.

For the aggressive normalized closed loop, an offline semidefinite search provides a common P and the verified endpoint norms $\|\Phi\|_P = 0.999939$ and $\|A_{\text{cl}}\|_P = 0.979743$. Therefore the segment bound in proposition 2 is 0.999939. This number certifies the linearized conditioned family; no SDP is solved during cycle execution.

4.3 Conditioned large-error study

The exact nonlinear map is simulated for 100 cycles from

$$\mathbf{x}_0 = [7.5143 \quad 20.8211 \quad 0.0314]^\top. \quad (28)$$

Three cases share this state: the conditioned aggressive law, the unconditioned conservative law, and open loop. Every interval is propagated with exact matrix exponentials rather than the linearized model. The initial normalized error, 0.983946, is intentionally well beyond the small perturbation used for local validation.

Without conditioning, the aggressive candidate reaches a minimum raw dwell duration of -43.045 μs , so direct application would invert an interval. The conditioner is active for 3 cycles, reaches $\beta_{\text{min}} = 0.293978$, and limits the maximum raw/applied offset from 65.217 to 19.172 μs . The minimum applied dwell is 3.000 μs , exactly the assumed bound. The conservative raw law remains feasible, with minimum dwell 15.502 μs , but crosses the 0.01 error threshold at cycle 20 rather than cycle 6.

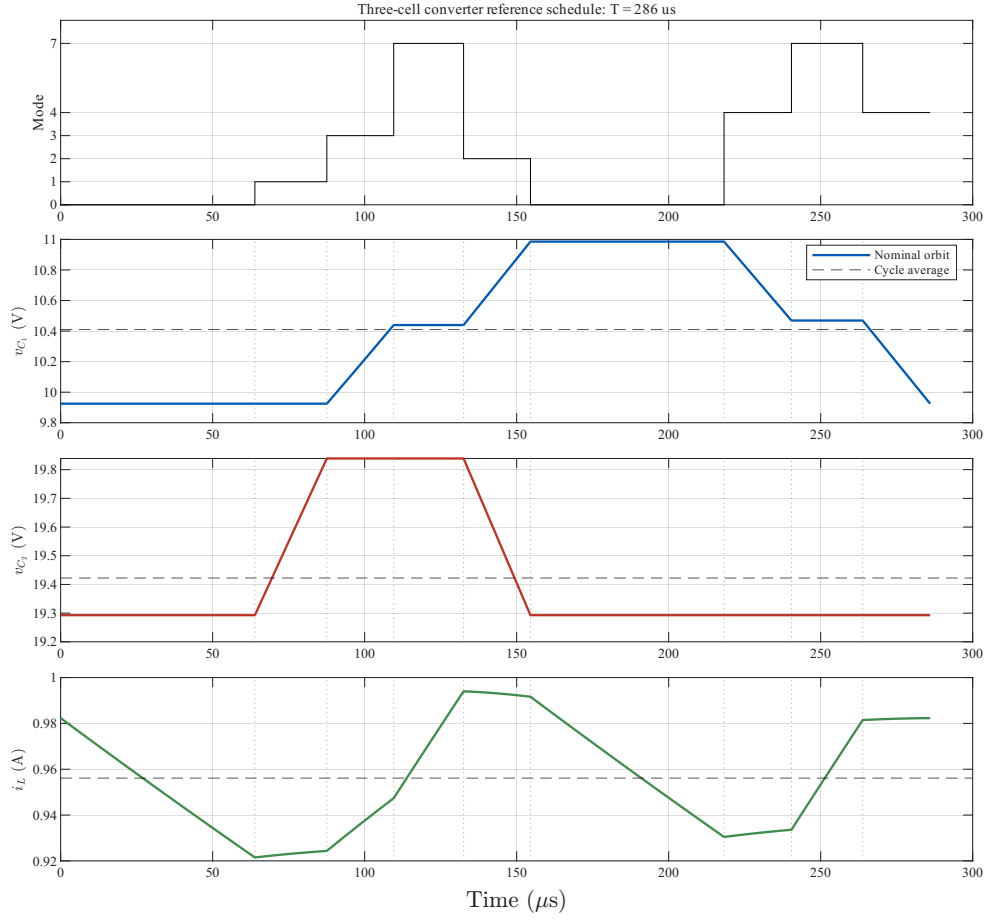


Figure 1: Three-cell converter reference schedule after closure reconciliation and the corresponding exact periodic trajectory. Dashed horizontal lines are cycle averages; dotted vertical lines are the eight movable interior switching instants.

Table 2: Validation, certificate, and exact nonlinear large-error results. Settling is the first cycle with $\|S_x^{-1}e_k\|_2 < 0.01$.

Quantity	Value
Relative errors: Φ ; Γ_τ	8.518e-12; 8.344e-11
Linearization residual slope	2.000
Spectral radii: open; conservative; aggressive	0.999850; 0.941468; 0.568151
Certified conditioned-family contraction bound	0.999939
Minimum conditioning factor; active cycles	0.293978; 3
Maximum raw; applied offset (μ s)	65.217; 19.172
Minimum raw; applied dwell (μ s)	-43.045; 3.000
Minimum conservative dwell (μ s)	15.502
Settling cycle: conditioned; conservative	6; 20
Final errors: conditioned; conservative; open	1.746e-15; 3.113e-06; 0.298010

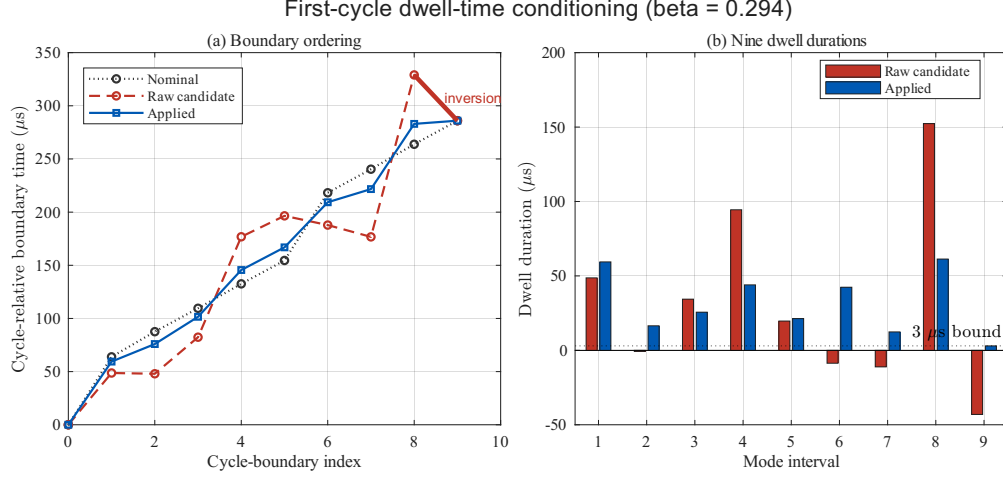


Figure 2: First-cycle operation of the dwell-time conditioner. (a) Nominal, raw-candidate, and applied cycle-boundary times. The highlighted downward raw segment is a boundary inversion; the conditioned schedule preserves ordering. (b) All nine raw and applied dwell durations. The raw candidate reaches $-43.045 \mu\text{s}$, whereas the applied schedule enforces the assumed $3 \mu\text{s}$ bound.

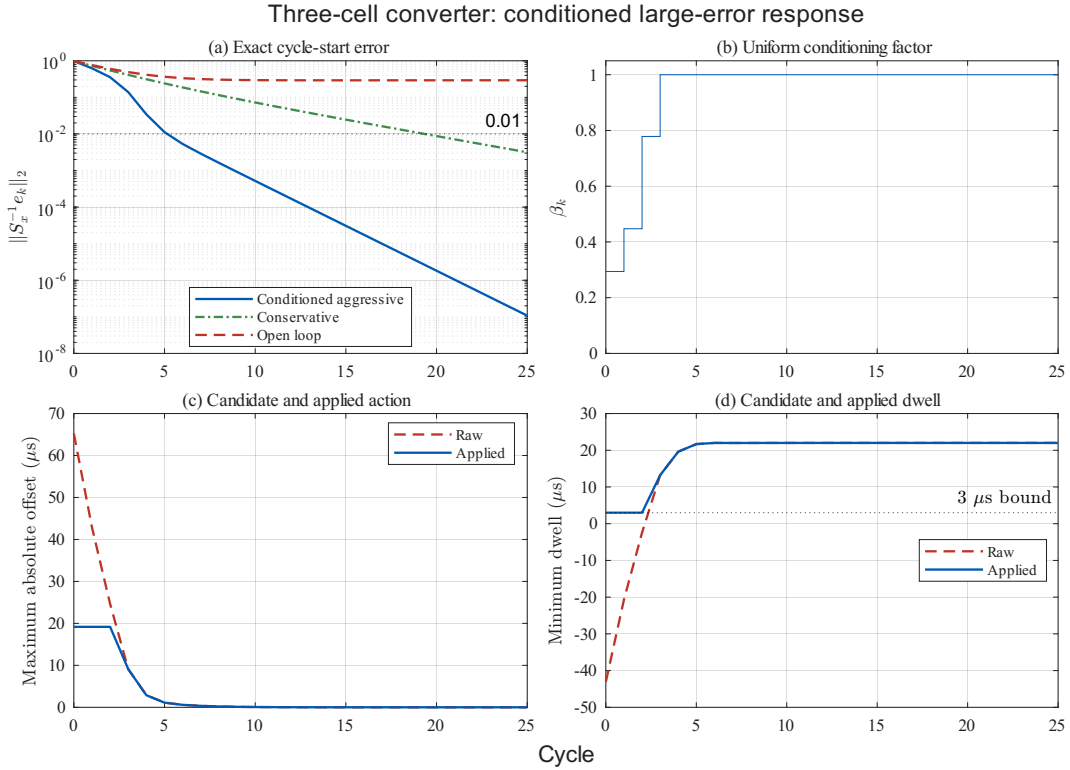


Figure 3: Exact nonlinear 100-cycle large-error response. (a) Normalized cycle-start errors for conditioned aggressive, unconditioned conservative, and open-loop cases. (b) Conditioning factor. (c) maximum absolute raw and applied switching-instant offsets. (d) minimum raw and applied dwell durations; the raw candidate becomes negative, whereas the applied schedule respects the assumed $3 \mu\text{s}$ bound.

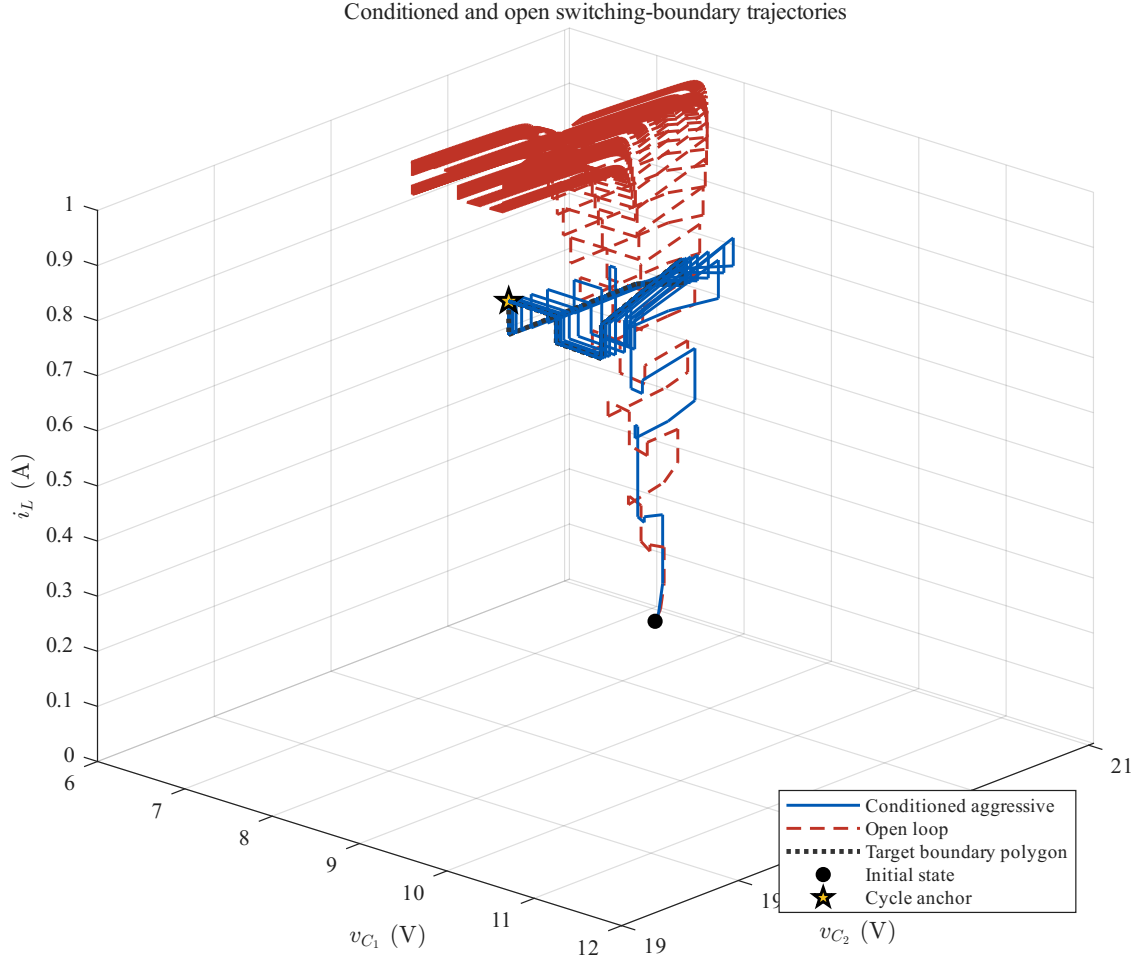


Figure 4: Exact switching-boundary trajectories from the large-error initial state. The conditioned aggressive response approaches the target boundary polygon and cycle anchor; the open-loop trajectory is shown for comparison. This visualization is simulation evidence beyond the local theorem, not a global nonlinear guarantee.

5 Discussion

The conditioner is an implementation-oriented fixed-period feasibility layer: it retains the largest feasible scalar multiple of a static timing action without changing its direction or the final cycle boundary. It does not replace constrained optimization when prediction or a different action direction is required. The common- P search is an offline certificate, not online workload.

Patiño et al. established the mode-dependent sensitivity [1], while Marcolino et al. supplied fixed-period coordinates, dwell inequalities, and LQR for switched actuators [2]. The derivation and checks here connect and validate those published foundations. The central addition is the radial conditioner and its common-Lyapunov analysis, not LQR synthesis.

The 3 μ s dwell bound is only a simulation assumption. Timer quantization, delays, dead time, hardware margin, estimation, uncertainty, and experiments are absent. Mode order, phase, and period are fixed, and exact nonlinear stability is proved only where conditioning is inactive. Proposition 2 applies to the linearized family; the large-error study is benchmark evidence, not a global nonlinear guarantee.

6 Conclusion

The maximal scalar conditioner enforces all minimum dwell durations in $O(N)$ time while preserving action direction and period. Endpoint induced- P norms certify every conditioned linearized matrix. The mode-dependent derivation, switched-actuator reduction, and direct validation support this result. Exact nonlinear simulation repaired an infeasible aggressive action and outpaced a feasible conservative law. Embedded evaluation must add timer quantization and hardware margins; no global nonlinear guarantee is made.

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